

## Math 421

Friday, December 5

## Announcements

- Homework 10 due **today** (last HW!)
- Final Exam –
  - Date: Monday, December 15<sup>th</sup>
  - Time: 5:05-7:05 pm
  - Location: Bardeen 140
- McBurney & Makeups
  - McBurney exams – Sign up through the Testing Center (RegisterBlast) early!
  - Makeup exams – If you emailed me about taking a makeup final – you should have received the date & time. If you did not, email me again.

# Upper & Lower Sums

**Definition.** Let  $f: [a, b] \rightarrow \mathbb{R}$  be bounded and  $P = \{t_0, t_1, \dots, t_{n-1}, t_n\}$  be a partition of  $[a, b]$ . The **upper and lower (Darboux) sums** for  $P$  are defined by

$$U(f, P) = \sum_{k=1}^n M_k (t_k - t_{k-1}) \quad M_k = \sup\{f(x) : x \in [t_{k-1}, t_k]\}$$

$$L(f, P) = \sum_{k=1}^n m_k (t_k - t_{k-1}) \quad m_k = \inf\{f(x) : x \in [t_{k-1}, t_k]\}$$

# Relationship Between Partitions

**Lemma 32.2.** If  $P, Q$  are partitions of  $[a, b]$  such that  $P \subseteq Q$  and  $f$  is bounded on  $[a, b]$ , then

$$L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P)$$

*Discussion.*

# Lower Sums $\leq$ Upper Sums

**Lemma 32.3.** Let  $P_1$  and  $P_2$  be partitions of  $[a, b]$  and let  $f$  be a function which is bounded on  $[a, b]$ . Then  $L(f, P_1) \leq U(f, P_2)$ .

Proof.

# Upper & Lower Integrals

**Definition.** Let  $f: [a, b] \rightarrow \mathbb{R}$  be bounded. The **upper (Darboux) integral** and the **lower (Darboux) integral** of  $f$  over  $[a, b]$  are defined by

- $U(f) = \inf\{U(f, P) : P \text{ is a partition of } [a, b]\}$
- $L(f) = \sup\{L(f, P) : P \text{ is a partition of } [a, b]\}$

$$L(f) \leq U(f)$$

**Theorem 32.4.** If  $f$  is a bounded function on  $[a, b]$ , then  $L(f) \leq U(f)$ .

Proof.

# Integrable

**Definition.** A function  $f$  which is bounded on  $[a, b]$  is **integrable** on  $[a, b]$  if

$$L(f) = U(f)$$

In this case, the value is called the **integral** of  $f$  and is denoted by

$$\int_a^b f = \int_a^b f(x) dx.$$

# Activity

For each of the following functions, the upper and lower integrals and determine if the function is integrable.

1. Define  $f(x) = c$ .

2. Define  $g(x) = \begin{cases} c, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$



# Equivalent Definition

**Thm.** (“Cauchy criterion for integrability”) If  $f$  is bounded on  $[a, b]$ , then  $f$  is integrable on  $[a, b]$  if and only if

for every  $\epsilon > 0$ , there exists a partition  $P$  of  $[a, b]$

such that  $U(f, P) - L(f, P) < \epsilon$

# Proof of Equivalence ( $\Rightarrow$ )

# Proof of Equivalence ( $\Leftrightarrow$ )

# Example

Let  $f(x)$  be defined as follows. Prove  $f$  is integrable on  $[-1,1]$ .

$$f(x) = \begin{cases} 0, & x \leq 0 \\ 1, & x > 0 \end{cases}$$

# Example

Prove, directly from the definition of upper and lower integrals, that  $f(x) = x$  is integrable on  $[0, b]$  and find the value of  $\int_0^b x \, dx$ .